

QEMU: For Faster Virtualisation And Better Emulation

ASHWIN GOPINATH

For the uninitiated, virtualisation and emulation might seem to be two words that can be used interchangeably, and that would be as wrong as you could get. Virtualisation involves simulating the functioning of a computer's hardware like CPU, CD-ROM, graphics controller and the like. While a program like VMWare can provide a virtual environment for running a virtual Windows PC, it cannot work on any real hardware other than its native x86 PC.

On the other hand, in an emulator, the virtual machine simulates the complete hardware in software. This allows an operating system of one computer architecture to be run on the architecture for which the emulator is written. If you have ever tried Contra on a PC, then you know what function an emulator performs.

Two birds with one sharp stone

In this month's DVD for *Electronics for You Plus* we bring to you QEMU, a software that can do both, emulate and virtualise.

QEMU, or Quick EMulator, is a virtual machine monitor that emulates CPUs through dynamic binary translation, providing a variety of models through which it runs a bevy of guest operating systems. Table I shows the various formats and architectures it supports.

How a virtual machine runs

All virtual machines generally work on the principle that the operating system allocates resources of the

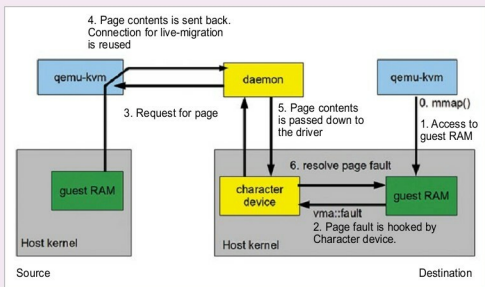


Fig. 1: QEMU on source and destination

QEMU software details

- QEMU as a whole is released under GNU general-public licence version 2
- Tiny Code Generator is released under BSD licence
- QEMU is a trademark of FabriceBellard
- Latest stable release 2.6.2 was launched on September 29, 2016
- Software type: Hypervisor
- Website: qemu.org

system to any process that requires it, be it disk space, memory, peripherals and so on. When a user program needs these resources, it sends a syscall to the operating system. When the syscall is sent to a real operating system, the operating system, in turn, accesses the hardware personally and facilitates the release.

However, in the case of a virtual machine, the guest operating system does not have access to the hardware. So it does something straight out of a con-job movie. It impersonates the host operating system. By using a process known as trap-and-execute, it traps syscalls and then

TABLE I
VARIOUS FORMATS
AND ARCHITECTURES
QEMU SUPPORTS

Architectures supported	Devices supported
IA-32 x86 PCs	CD-ROM
X86-64 PCs	ATA/SATA controller
MIPS R6	Graphics card
SPARC Sun 4m/4u	Parallel port
SH4 SHIX	PC Speaker
Power PC	SCSI controller
ETRAX CRIS	Serial interface

reroutes these via the guest operating system to masquerade as coming from the host operating system. This